

Exact certificates for the first two cases not covered by Steinberger's cyclotomic minimum-degree theorem

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27 August 2026

Internal candidate manuscript, version 0.1-beta. The mathematical proof and complete Markdown manuscript have passed project-internal zero-trust audits. External mathematical review and formal peer review are pending.

Abstract

Let $\Phi_n(x)$ be the n -th cyclotomic polynomial. Steinberger asked for the lowest degree of a nonzero polynomial with nonnegative coefficients divisible by Φ_n and conjectured that the unique monic minimizer is the regular p -gon polynomial

$$1 + x^{n/p} + \dots + x^{(p-1)n/p},$$

where p is the smallest prime divisor of n . His general theorem leaves $46189 = 11 \cdot 13 \cdot 17 \cdot 19$ and $96577 = 13 \cdot 17 \cdot 19 \cdot 23$ as the first two uncovered values. We prove the conjecture for both values. The proof is computer assisted but exact: for each parameter we provide an integer element of the orthogonal complement of the cyclotomic-array space which vanishes on the regular polygon and has a strict sign at every other position up to the proposed boundary degree. The certificates are checked by two exact routes, one using CRT tensor marginals and one expanding the full multidimensional array and verifying every fiber sum. Consequently the minimum degrees are 41990 and 89148, respectively, and the monic minimizers over the nonnegative reals are unique. These are two fixed-parameter results, not a proof of the general conjecture.

1 Introduction

For a positive integer n , consider the optimization problem

$$\min\{\deg f : 0 \neq f \in \mathbb{R}_{\geq 0}[x], \Phi_n(x) \mid f(x)\}. \quad (1.1)$$

If p is the smallest prime divisor of n , the polynomial

$$R_{n,p}(x) = 1 + x^{n/p} + \dots + x^{(p-1)n/p} \quad (1.2)$$

is divisible by $\Phi_n(x)$ and has degree $(p-1)n/p$. Steinberger [1] conjectured that this is always the minimum degree and that $R_{n,p}$ is the unique monic minimizer. His Theorem 1 proves the conjecture when n is even, when n is a prime power, and, for squarefree notation $n = pq_1 \cdots q_k$, whenever

$$\frac{2}{p} > \frac{1}{q_1} + \dots + \frac{1}{q_k}. \quad (1.3)$$

The same paper identifies

$$46189 = 11 \cdot 13 \cdot 17 \cdot 19, \quad 96577 = 13 \cdot 17 \cdot 19 \cdot 23 \quad (1.4)$$

as the first two values not covered by that theorem. It also explains that each fixed case is a finite linear-programming problem, while noting that the resulting systems were too large for the computations carried out there.

The purpose of this note is to settle exactly these two fixed cases. Our main result is the following.

Theorem 1.1. *Let $f \in \mathbb{R}_{\geq 0}[x]$ be nonzero.*

1. *If $\Phi_{46189}(x) \mid f(x)$, then $\deg f \geq 41990$. Equality holds precisely for positive scalar multiples of $1 + x^{4199} + \dots + x^{10 \cdot 4199}$.*
2. *If $\Phi_{96577}(x) \mid f(x)$, then $\deg f \geq 89148$. Equality holds precisely for positive scalar multiples of $1 + x^{7429} + \dots + x^{12 \cdot 7429}$.*

In particular, in each case the displayed polynomial is the unique monic minimizer.

The mathematical reduction is Steinberger's separation criterion [1, Lemma 1]. The new finite objects are two exact integer certificates. Floating point linear programming was used to discover them, but it is not part of the proof: the final claims depend only on the integer files and exact arithmetic.

Section 2 recalls the separation criterion and makes the coefficient field in the uniqueness statement explicit. Section 3 gives the CRT-array representation and the certificate data. Section 4 describes the exact verification. Section 5 records how the larger certificate was found, while separating that history from the proof. Section 6 states the scope relative to prior work.

2 The separation criterion

We use Steinberger's notation in a form adapted to exact certificates. Let n be squarefree, let

$$n = p_1 p_2 \cdots p_s, \quad p_1 = p < p_2 < \cdots < p_s,$$

and put

$$P = \frac{n}{p}, \quad d = (p-1)P.$$

Let $V_n \subseteq \mathbb{R}^n$ be the space of coefficient vectors of polynomials of degree less than n which are divisible by Φ_n . Define

$$H = \{0, P, 2P, \dots, (p-1)P\} \quad (2.1)$$

and

$$F^+ = \{0, 1, \dots, d-1\} \setminus H. \quad (2.2)$$

Thus H is the support of the regular p -gon polynomial, including its degree- d endpoint. Steinberger's Lemma 1 uses a vector in $V_n^\perp$ which is zero on H and strictly positive on F^+ . We use the negative of that vector.

Proposition 2.1 (Steinberger's criterion, sign reversed). *Suppose there is a vector $w \in V_n^\perp$ such that*

$$w_j = 0 \quad (j \in H), \quad w_j < 0 \quad (j \in F^+). \quad (2.3)$$

Then every nonzero $f \in \mathbb{R}_{\geq 0}[x]$ divisible by Φ_n and of degree at most d is supported on H .

Proof. Pad the coefficient vector a of f with zeros to length n . Because $\deg f \leq d < n$ and $\Phi_n \mid f$, we have $a \in V_n$. Hence

$$0 = \langle w, a \rangle = \sum_{j=0}^d w_j a_j.$$

Every summand is nonpositive, and every summand indexed by F^+ is strictly negative whenever the corresponding coefficient is positive. Therefore $a_j = 0$ for every $j \in F^+$, so the support is contained in H . $\square$

For completeness, we record the step which turns support containment into the real-coefficient uniqueness statement. It is important here not to appeal merely to irreducibility over $\mathbb{Q}$, since our coefficients lie in $\mathbb{R}$.

Lemma 2.2 (real-coefficient boundary rigidity). *Let n be squarefree, let p be its smallest prime divisor, and put $P = n/p$. Suppose $0 \neq f \in \mathbb{R}_{\geq 0}[x]$, $\Phi_n \mid f$, $\deg f \leq (p-1)P$, and $\text{supp}(f) \subseteq H$. Then*

$$f(x) = c(1 + x^P + \cdots + x^{(p-1)P}) \quad (2.4)$$

for some $c > 0$.

Proof. Write $f(x) = g(x^P)$ with $g \in \mathbb{R}[x]$ and $\deg g \leq p-1$. Let ζ_n be a primitive n -th root of unity. For every $a \in (\mathbb{Z}/n\mathbb{Z})^*$,

$$0 = f(\zeta_n^a) = g((\zeta_n^P)^a).$$

The root ζ_n^P is primitive of order p . Because n is squarefree, the reduction map

$$(\mathbb{Z}/n\mathbb{Z})^* \longrightarrow (\mathbb{Z}/p\mathbb{Z})^* \quad (2.5)$$

is surjective by the Chinese remainder theorem. Consequently g vanishes at every primitive p -th root. Therefore $\Phi_p \mid g$ in $\mathbb{R}[x]$. Since both $\deg g \leq p-1$ and $\deg \Phi_p = p-1$, we have $g = c\Phi_p$ for a real scalar c . Nonzero nonnegative coefficients force $c > 0$, proving (2.4). $\square$

Proposition 2.1 and Lemma 2.2 show that a single boundary certificate proves both the degree lower bound and boundary uniqueness. Indeed, a polynomial of degree less than d would be forced to equal a positive scalar multiple of the regular polygon, which has degree exactly d .

3 CRT arrays and the two certificates

We recall the exact coordinate model used to store the certificates. For $n = p_1 \cdots p_s$, define

$$\psi(t_1, \dots, t_s) = \sum_{r=1}^s t_r \frac{n}{p_r} \pmod{n}, \quad 0 \leq t_r < p_r. \quad (3.1)$$

Every factor n/p_r is invertible modulo p_r , so ψ is a bijection. Its inverse sends an exponent j to

$$t_r(j) = j \left(\frac{n}{p_r} \right)^{-1} \pmod{p_r}. \quad (3.2)$$

Under this identification, the space V_n is spanned by indicators of fibers parallel to the coordinate axes [1]. It follows that

$$V_n^\perp = \mathbf{1}_{p_1}^\perp \otimes \cdots \otimes \mathbf{1}_{p_s}^\perp. \quad (3.3)$$

For $1 \leq i < p_r$, let $b_i^{(r)} = e_i - e_0$. These vectors form an integer basis of $\mathbf{1}_{p_r}^\perp$. An integer coordinate array

$$U = (U_{i_1, \dots, i_s}), \quad 1 \leq i_r < p_r,$$

therefore determines

$$W = \sum_{i_1=1}^{p_1-1} \cdots \sum_{i_s=1}^{p_s-1} U_{i_1, \dots, i_s} b_{i_1}^{(1)} \otimes \cdots \otimes b_{i_s}^{(s)} \in V_n^\perp. \quad (3.4)$$

The certificate value at exponent j is

$$w_j = W_{t_1(j), \dots, t_s(j)}. \quad (3.5)$$

The two retained boundary certificates have the following exact summaries.

n	prime factors	$P = n/p$	logical shape of U	$\min_{F^+} w_j$	$\max_{F^+} w_j$
46189	11, 13, 17, 19	4199	$10 \times 12 \times 16 \times 18$	-4272093684	-999961
96577	13, 17, 19, 23	7429	$12 \times 16 \times 18 \times 22$	-40866441870	-999967

For $n = 46189$, all 11 values on

$$H = \{0, 4199, \dots, 10 \cdot 4199\}$$

are zero, and all other 41,980 values through degree 41,990 are negative. For $n = 96577$, all 13 values on

$$H = \{0, 7429, \dots, 12 \cdot 7429\}$$

are zero, and all other 89,136 values through degree 89,148 are negative.

The certificate files are anchored by SHA-256:

n	file	SHA-256
46189	boundary-uniqueness-46189.npz	79d9042e4c050a7a2f5f12b95f47b2c53857b55a531419ece063fb6d6dc051b4
96577	boundary-uniqueness-96577.npz	78a03fa247e2d33cd232e7d0365c767c106feb8895ea55e16ababb980ce4a563

The NPZ files store the integer coordinates as flat vectors; the verification programs reshape them to the logical tensor dimensions displayed above.

Proof of Theorem 1.1. For each of the two values of n , exact verification establishes $W \in V_n^\perp$ for the reconstructed array, together with zero values on H and strict negativity on F^+ . Proposition 2.1 forces every nonnegative multiple of degree at most the proposed boundary to be supported on H . Lemma 2.2 then shows that it is a positive scalar multiple of the corresponding regular polygon. This proves the lower bound, characterizes equality, and gives monic uniqueness. $\square$

4 Exact verification

The proof objects are the final integer arrays, not the linear-programming transcripts. We provide two exact evaluation routes.

4.1 Tensor-marginal evaluation

Formula (3.4) can be evaluated without materializing all n entries. For $t = (t_1, \dots, t_s)$, let $Z(t) = \{r : t_r = 0\}$. A nonzero coordinate t_r forces $i_r = t_r$, while a zero coordinate contributes a factor -1 and a sum over i_r . Thus

$$W_t = (-1)^{|Z(t)|} \sum_{\substack{1 \leq i_r < p_r \ (r \in Z(t)) \\ i_r = t_r \ (r \notin Z(t))}} U_{i_1, \dots, i_s}. \quad (4.1)$$

The marginal verifiers use (3.2) and (4.1) to recompute every exponent through the boundary using exact integer arithmetic. They check n , the prime tuple, the boundary, the support H , and the coordinate count, and recompute the certificate hash for comparison with the accompanying metadata. The fixed hashes in Section 3 are the manuscript-level anchors.

4.2 Full-array verification

The second route explicitly expands (3.4) to arrays of shapes

$$11 \times 13 \times 17 \times 19 \quad \text{and} \quad 13 \times 17 \times 19 \times 23.$$

It then checks that every fiber sum in each of the four coordinate directions is exactly zero. This verifies membership in $V_n^\perp$ without calling the marginal evaluation routine. Finally it applies the CRT map to all boundary positions and checks the required zero set and strict sign.

All sums are exact integers. A deliberately conservative overflow audit bounds the largest possible fiber accumulation for the three retained certificates by 4.95×10^{16} , below the signed 64-bit limit 9.22×10^{18} . An additional audit expanded the arrays with arbitrary precision Python integers and reproduced the same extrema, counts, zero sets, and fiber sums.

For $n = 46189$ we also retain a logically redundant strict-separation certificate for all degrees at most 41,989. Its SHA-256 is

9f0e34517f3ae92161e359e0345563d1ae4ff73a5cdc99693c0e82657dce0e80.

It provides a separate check of the minimum-degree conclusion, although the boundary certificate alone already proves the full statement.

5 Certificate discovery and integerization

This section records how the proof objects were found. None of the numerical statuses described here is used as a theorem premise.

An element of $V_n^\perp$ was parametrized by its integer-basis coordinates U . The boundary problem imposes p homogeneous equalities on H and strict inequalities on every position in F^+ . In floating-point discovery these were normalized to inequalities of the form $w_j \leq -1$.

For $n = 46189$, a direct sparse formulation produced a feasible point which could be scaled, rounded, and repaired exactly. For $n = 96577$, the recorded discovery run of the original 76,032-variable formulation reached its time limit without producing a proof object. This was a solver outcome, not a mathematical conclusion.

The successful reformulation eliminated 12 independent boundary equalities using a pivot minor of determinant $+1$. The recorded reduced bounded interior-point run then produced a strict floating-point separator. After scaling by 10^6 and rounding, the boundary equalities were repaired over the integers using a 12-by-12 minor of determinant -1 . The resulting integer vector was then discarded as a numerical candidate unless and until the exact verifiers accepted it.

Metadata fields recording the floating-point scale or repair size describe this discovery history only. They are not proof evidence and need not be reconstructed from the final NPZ files.

6 Scope and relation to prior work

Steinberger [1] formulated the optimization problem, proposed the general conjecture, established the separation criterion used here, and proved the large parameter range summarized by (1.3). The same paper explicitly lists 46189 and 96577 as the first two values outside that theorem. The present note contributes exact certificates and their verification for those two fixed values.

The result does not prove the conjecture for all squarefree integers with four prime factors, does not provide an infinite parameter family, and does not settle the next listed value 215441. The exact computation also does not by itself explain why a separator should exist in general.

A targeted search completed on 27 August 2026 found no public paper, preprint, author update, or indexed computational archive directly treating either of these two certificates. This supports writing the fixed-parameter result but is not an assertion of exhaustive global priority. Unindexed manuscripts and independent unpublished computations remain possible.

7 Reproducibility

The exact certificate files, metadata, and verification programs are retained in the project `verification` directory. With Python, NumPy, and these files available, the principal checks

are:

```
& '.\venv\Scripts\python.exe' '
loops\CYCLOTOMIC-MINDEG-0001\verification\verify_boundary_uniqueness.py' '
--certificate 'loops\CYCLOTOMIC-MINDEG-0001\verification\results\boundary-uniqueness-46189.npz' '
--metadata 'loops\CYCLOTOMIC-MINDEG-0001\verification\results\boundary-uniqueness-46189.json'

& '.\venv\Scripts\python.exe' '
loops\CYCLOTOMIC-MINDEG-0001\verification\verify_full_array.py' '
--mode boundary '
--certificate 'loops\CYCLOTOMIC-MINDEG-0001\verification\results\boundary-uniqueness-46189.npz' '
--metadata 'loops\CYCLOTOMIC-MINDEG-0001\verification\results\boundary-uniqueness-46189.json'

& '.\venv\Scripts\python.exe' '
loops\CYCLOTOMIC-MINDEG-0001\verification\verify_boundary_96577.py' '
--certificate 'loops\CYCLOTOMIC-MINDEG-0001\verification\results\boundary-uniqueness-96577.npz' '
--metadata 'loops\CYCLOTOMIC-MINDEG-0001\verification\results\boundary-uniqueness-96577.json'

& '.\venv\Scripts\python.exe' '
loops\CYCLOTOMIC-MINDEG-0001\verification\verify_full_array_96577.py' '
--certificate 'loops\CYCLOTOMIC-MINDEG-0001\verification\results\boundary-uniqueness-96577.npz' '
--metadata 'loops\CYCLOTOMIC-MINDEG-0001\verification\results\boundary-uniqueness-96577.json'
```

The current project environment used Python 3.13.11 and NumPy 2.5.2. SciPy 1.17.0 and HiGHS were used during discovery, but are not required by the mathematical verification of the final integer arrays.

Acknowledgement of AI assistance

The author used OpenAI Codex as an AI-assisted tool for literature discovery, computational exploration, certificate construction, proof stress testing, and language editing. The author reviewed the mathematical argument and assumes responsibility for the claims and presentation.

References

- [1] J. P. Steinberger, *The lowest-degree polynomial with nonnegative coefficients divisible by the n -th cyclotomic polynomial*, Electronic Journal of Combinatorics **19** (2012), no. 4, Paper P1.
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